PROJECT BRIEF

# The AGI Model v2.1

A Transparent Replacement for Kenya's SHA Healthcare Means-Testing Algorithm

| 29.8M | KES 52B | 47 |
|---|---|---|
| Kenyans affected | annual revenue opportunity | counties — national scope |

# 1. Executive Overview

Kenya's Social Health Authority (SHA) uses an artificial intelligence algorithm to decide how much each of 29.8 million registered citizens must pay for healthcare. The algorithm is broken. It overcharges the poorest households, undercharges the wealthy, and cannot explain its own decisions — and SHA's own evaluator warned it was biased before it was deployed.

The AGI (Adjustable Gross Income) Model v2.1 is a complete, working replacement. It was independently designed, built, and tested over the past academic year by a KCA University student, and it directly fixes the specific failures that independent journalists, government auditors, and Kenya's courts have all separately documented.

This is not a theoretical proposal. The research has already been filed as formal evidence with the Commission on Administrative Justice, submitted to Kenya's national policy research institute (KIPPRA), and sent directly to SHA's Chief Executive Officer. A working software prototype exists and runs today.

This brief explains the problem, the solution, who it helps and where, the path to government adoption, and — specifically for this office — why this is a viable venture with a real, generalizable market, not only a single-government policy submission.

**The one-sentence version**

A government healthcare algorithm is provably broken and is currently under three separate legal challenges; an independent student has already built, tested, and legally filed the fix — and the underlying technology and approach has a market far beyond Kenya's SHA alone.

# 2. The Problem

## 2.1 What SHA Is and Why It Matters

In October 2024, Kenya replaced the National Hospital Insurance Fund (NHIF) with the Social Health Authority (SHA) as part of a national push toward Universal Health Coverage. The promise was that no Kenyan family would again need to run a harambee — a community fundraiser — to pay a hospital bill.

Because roughly 80% of Kenya's workforce operates in the informal sector with no formal payslip, SHA cannot simply tax a salary. Instead, it uses a machine-learning algorithm — officially called the Means Testing Instrument (MTI), built on a statistical method known as Proxy Means Testing (PMT) — to estimate each household's income from indirect signals: housing materials, asset ownership, utility access, and household composition. That estimate determines a mandatory monthly premium, currently set at 2.75% of estimated income with a KES 300 minimum.

## 2.2 The Algorithm Is Demonstrably Biased

The MTI was trained on the 2021 Kenya Continuous Household Survey — data that predates the post-pandemic cost-of-living crisis. SHA's own contracted evaluator, the development consultancy IDinsight, flagged the model as inequitable and structurally outdated before it was deployed. The government deployed it anyway, under pressure to maximise revenue.

An independent technical audit — conducted by the investigative outlets Africa Uncensored, Lighthouse Reports, and The Guardian after a freedom-of-information dispute that required intervention by the Commission on Administrative Justice — reconstructed the algorithm and quantified the damage:

| 80% | 56.02% | 65.22% |
|---|---|---|
| of the poorest households have their income overestimated | exclusion error for poor households with basic electricity access | exclusion error for poor female-headed households (vs 34.12% for male-headed) |

In practice, this means an unemployed mother with no income can be assessed a premium of KES 650–800 per month — 10 to 20% of her actual income — because her household owns a radio or has basic electrical access. The automated appeals process frequently rejects challenges without written justification, and the algorithm's internal weightings have never been fully disclosed to the public.

### 2.3 The Financial Crisis This Has Created

The consequence of an algorithm the public does not trust is that people stop paying. Of 29.8 million registered Kenyans, only about 5 million pay their premiums regularly. SHA's own data, presented to Parliament, shows the informal sector loss ratio rose from 1,922% in January 2025 to 2,655% by mid-2025 — meaning SHA pays out more than KES 26 in claims for every KES 1 it collects from informal sector members.

The downstream effects are severe: 92% of private healthcare providers report serious financial distress from delayed reimbursements, claim processing now routinely takes 90–120 days, and over KES 11 billion in fraudulent claims was identified by government audit in a single six-month period. The National Assembly's Departmental Committee on Health publicly declared the system “unsustainable” in March 2026.

This crisis has also become more urgent because of events outside Kenya's control: in early 2025, US funding bodies (USAID and PEPFAR) suspended health funding that had previously covered the majority of Kenya's HIV and broader health programme costs. Domestic revenue generation through SHA is no longer simply desirable — it is now close to the only available funding mechanism left.

### 2.4 The Legal Pressure Is Mounting Quickly

SHA is currently the subject of overlapping legal challenges, each independently confirming that the current system cannot continue as it stands:

- **CAJ disclosure order —** The Commission on Administrative Justice (Order CAJ/2026/05/0847) ordered SHA to fully disclose its algorithm within 21 days. SHA partially complied in November 2025 — releasing the 43 variables used but withholding the actual weightings — and has still not satisfied the order in full.
- **High Court ruling —** In March 2026, the High Court (Justice Bahati Mwamuye) ruled the SHA rollout unconstitutional, finding the scheme was launched before adequate administrative and technological infrastructure existed, and gave the government a strict 90-day window — expiring June 17, 2026 — to remedy the gaps.
- **Awino petition —** A separate constitutional petition (Awino v. SHA and others) was filed in May 2026 arguing SHA conducts insurance-like functions without proper statutory authority. In June 2026 it was transferred to the Constitutional and Human Rights Division in Nairobi specifically because it overlaps with other pending constitutional petitions against SHA — evidence that this is now a recognised pattern of legal exposure, not an isolated dispute.
- **Mwita ruling —** In June 2025, a separate ruling found that applying the 2.75% SHIF levy to gross income constitutes unconstitutional double taxation, since the same income is already taxed once under the Income Tax Act.

Three of Kenya's constitutional oversight mechanisms — an administrative ombudsman, the High Court, and a constitutional petition cluster — are simultaneously active on this single algorithm. That is an unusually concentrated legal and political opening.

## 3. Who This Affects, and Where

This is not a narrow technical problem. It touches nearly every Kenyan household and every county in the country.

### Who is affected

- 24.8 million registered but non-paying informal sector Kenyans — the population that left the system because of unaffordable, unexplained premiums
- Female-headed households, who face nearly double the exclusion error rate of male-headed households
- Rural and ASAL (arid and semi-arid land) households, where the algorithm's accuracy is documented to be weakest
- Casual labourers and the unemployed, who are billed premiums based on static assets rather than actual ability to pay
- Private healthcare providers nationwide, who are owed billions in delayed reimbursements as a direct consequence of the compliance collapse

### Where

- All 47 counties — SHA is a national system
- 23 specifically classified ASAL counties, where the current algorithm's rural sub-model performs worst (independently measured accuracy of $R^2$=0.46, versus $R^2$=0.66 in urban areas)
- Disproportionately concentrated in informal settlements and rural counties, where Mchanga — Kenya's largest medical crowdfunding platform — has recorded over 53,000 medical fundraisers in the past year, the exact “harambee for hospital bills” problem SHA was created to eliminate

## 4. The Solution — The AGI Model v2.1

The Adjustable Gross Income (AGI) Model replaces SHA's opaque black-box algorithm with a transparent, auditable, and constitutionally defensible calculation. Every design decision in the model responds directly to a specific, documented failure of the current system.

| Current MTI failure | AGI model fix |
|---|---|
| Black-box regression; citizens cannot see why they were charged | SHAP deduction receipt for every household — a plain-language, itemised breakdown of exactly how the premium was calculated |
| Trained on stale 2021 survey data | Calibrated to current M-Pesa behavioural data and the 2024 FinAccess Survey; scheduled to retrain on the 2025/26 KIHBS national survey at zero additional cost |
| Applies levy to gross income (ruled unconstitutional double taxation) | Applies the levy to Adjusted Gross Income — after legitimate deductions — directly resolving the June 2025 court ruling |
| No correction for rural / ASAL accuracy gap | Location-weighted multipliers specific to Kenya's 23 ASAL counties |
| Misclassifies women's M-Pesa transaction patterns as personal income | Behavioural proxy correction recognising school-fee, remittance, and household-management transaction patterns |
| SMS-based verification fails for rural and elderly patients without smartphones | *147# USSD pathway works on any phone, on any network, even offline |

The model is fully documented — 34 published coefficients, complete algorithm pseudocode, a legal and constitutional compliance analysis, a Data Protection Act-compliant privacy framework, and an actuarial financial model. A working software prototype (built in React, simulating all 47 counties) demonstrates the algorithm end-to-end.

## 5. Why This Matters — The Benefits

### 5.1 For the Government

— An estimated KES 52 billion increase in annual SHA revenue (from approximately KES 87B to KES 139B in contributions) by restoring informal-sector trust and compliance — directly addressing the 2,655% loss ratio crisis

— Simultaneous resolution of three active legal exposures: the CAJ disclosure order, the constitutional double-taxation finding, and the AI Bill 2026's explainability requirement once enacted

— A working fix for the rural and elderly access barrier created by SHA's own SMS verification system — already publicly documented as a cause of denied care

### 5.2 For Citizens

— A premium calculation every household can actually read and understand, replacing an opaque number with a written explanation

— Fairer outcomes specifically for the groups most harmed today: the poorest quintile, female-headed households, and rural/ASAL populations

— A functioning appeals pathway grounded in a documented, auditable calculation rather than an unexplained rejection

### 5.3 For Healthcare Providers

- A larger, more stable contributor base reduces the adverse-selection pressure currently driving provider reimbursement delays
- Additional SHA revenue creates fiscal room to address the 90–120 day claims backlog

## 6. The Business Case — Why This Is More Than a Policy Submission

This section is specifically for this office. The research began as an academic project addressing a single Kenyan government system. What it has revealed is a generalizable opportunity: algorithmic accountability technology for government social-protection systems, in a market that is structurally guaranteed to grow.

### 6.1 The Market Is Larger Than SHA

Proxy Means Testing is not unique to Kenya — SHA itself has publicly cited Colombia (Sisben) and Indonesia (DTKS) as comparable systems. Variations of the same approach are used across dozens of low- and middle-income countries to allocate social protection, subsidies, and health financing. Every one of those systems faces the same structural tension SHA does: governments need a low-cost way to estimate informal-sector income, and every existing approach trades away either accuracy, fairness, or transparency to get it.

Within Kenya alone, the same structural problem — an opaque, publicly funded, unaccountable algorithm — has been separately identified in at least three other government systems: the Affordable Housing Levy, NSSF contribution assessments, and KRA/eTIMS compliance gaps. The underlying architecture built for SHA (transparent income estimation with explainable, auditable outputs) is directly portable to each of these.

Kenya's AI Bill 2026, currently before the Senate, will impose explainability and fairness-audit requirements on exactly this category of government algorithm once enacted. That creates a compliance market on a fixed legislative timeline — government agencies will need exactly this kind of audit and remediation capability, on a deadline they do not control.

### 6.2 Possible Revenue Models

| Model | Description |
|---|---|
| Government licensing & implementation | Direct contract with SHA (or a similar agency) to license, integrate, and maintain the AGI model as a replacement or parallel system |
| Algorithmic compliance auditing | A recurring audit-as-a-service offering for public-sector algorithms required to demonstrate AI Bill 2026 compliance — a fixed, government-mandated demand curve |
| Open-core / enterprise model | Core algorithm released open-source (already CC-BY 4.0 licensed) for credibility and adoption; paid enterprise tier for real-time fraud detection, county-level dashboards, and integration support |
| Cross-border technical assistance | Advisory and technical implementation support to other governments and development partners (World Bank, Gates Foundation, Global Fund) facing the same PMT reform pressure |

### 6.3 Early Competitive Position

- First documented, working, transparent replacement for a major African government means-testing algorithm — built and filed before any competing proposal
- Deep, citable domain documentation: legal analysis, financial model, and direct engagement with the government oversight bodies (CAJ, SHA, KIPPRA) already underway

- — Built-in regulatory alignment with Kenya's AI Bill 2026 from inception, rather than retrofitted after enactment
- — Open-source licensing (CC-BY 4.0) establishes public authorship and trust while preserving the option for a commercial enterprise layer

### 6.4 Honest Risk Factors

- — Government sales cycles are slow and procurement is politically sensitive; the proposed path (see Section 7) is designed specifically to minimise this risk through a zero-commitment pilot rather than an immediate full replacement ask
- — Some of the model's most powerful features depend on a future data-sharing agreement with Safaricom (M-Pesa); this is identified as a Phase 0 dependency, not an assumption, and the model has a documented fallback if that agreement is not secured
- — This remains, at this stage, a single-founder research project; institutional backing, mentorship, and potentially seed support materially de-risk the next 6–12 months

## 7. The Path to Government Backing

The strategy is deliberately staged to avoid asking any institution — SHA, KCA, or otherwise — to take on more risk than necessary at each step.

| Stage | What happens |
|---|---|
| 1. Academic validation | KCA University endorsement; co-authored policy brief submitted to KIPPRA (Kenya Institute for Public Policy Research and Analysis) for independent review and possible publication |
| 2. Parliamentary entry point | Submission to the National Assembly Departmental Committee on Health, which has already publicly declared SHA “unsustainable” and is actively seeking solutions |
| 3. Zero-commitment parallel run | SHA runs the AGI model silently alongside its current system on live data for a defined pilot period — no replacement commitment, only a side-by-side accuracy and fairness comparison |
| 4. Procurement, only if justified by data | Formal integration proceeds only once the parallel-run data demonstrates measurable improvement — evidence-led, not advocacy-led, adoption |

This pathway is already in motion. A citizen submission has been filed with the Commission on Administrative Justice, supplementary evidence has followed, and direct outreach has been sent to SHA's CEO, KIPPRA's leadership, and the international investigative outlets (Lighthouse Reports, The Guardian) whose reporting first exposed the algorithm's bias.

## 8. Traction So Far

| 20+ | 1 | 5 |
|---|---|---|
| research documents completed | working MVP — React, 47 counties | formal outreach submissions sent |

- — **Filed —** Citizen submission filed with the Commission on Administrative Justice under Order CAJ/2026/05/0847, including a complete algorithm disclosure and five worked household explanation examples — going further than SHA's own partial November 2025 disclosure
- — **Submitted —** Formal submission sent to KIPPRA requesting independent policy review and consideration for publication as a national Policy Brief
- — **Sent —** Direct submission sent to SHA's Chief Executive Officer proposing a zero-commitment 90-day parallel run

— **Sent —** Outreach sent to Lighthouse Reports and The Guardian, the investigative outlets whose reporting first forced SHA's partial algorithm disclosure
— **Built —** A complete, working MVP demonstrating the algorithm across all 47 counties, with full financial modelling and SHAP explainability built in

## 9. What We Are Asking From Business Incubation

**Four specific requests**

— **1.** Incubation support — workspace, mentorship, and technical guidance to continue developing the platform and formalise it as a venture over the coming months
— **2.** Institutional backing — a letter of support strengthening the existing CAJ and KIPPRA submissions, and lending institutional credibility to direct government engagement
— **3.** Introductions — to any existing govtech, public-sector innovation, or policy networks the incubator has relationships with, particularly around the National Assembly Health Committee and Ministry of Health
— **4.** Guidance on formalising the business model — specifically, help thinking through legal structure, government contracting realities, and how to sequence the licensing, compliance-auditing, and cross-border advisory revenue paths described in Section 6

None of these requests require the incubator to take an immediate position on SHA itself. They are requests to help a Kenyan-built, already-in-motion venture move from independent research to a properly structured, mentored, and resourced project — one that already has a working product, a filed legal record, and active engagement with three separate national institutions.